

Proposed Solution

Date: 30 September 2026

Team ID: SWTID-2026-3656

Project Name: EduGenie – Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Project Overview

Objective: Provide a simple AI-powered learning assistant for students.

Scope: Five focused learning tools in one responsive browser interface.

Core design: FastAPI routes call modular Python functions; the frontend uses `fetch()` to update results in place.

Feature Workflow

1. Student chooses a card.
2. Student enters a concept, question, topic, or passage.
3. Frontend posts form data to the matching endpoint.
4. Backend validates input.
5. Local or Gemini model generates content.
6. Frontend displays the result and error states safely.

Key Design Decisions

- Use `AutoModelForSeq2SeqLM`, not `pipeline("text-generation")`, for the T5 model.
- Cache the local model with `lru_cache`.
- Load `.env` before importing modules that inspect the key.
- Validate quiz JSON before displaying it.
- Keep the frontend framework-free for easy evaluation.